Networking concepts, DNS, IP, Subnets, & Ports

- Task 1: DNS – How Names Become IPs?
    - The DNS is a database of domain name and IP address information that enables computers to locate the correct IP address for a hostname of URL input into it.
  - Explain in 3-4 line: What happens when you type google.com in your browser?
    - The DNS is responsible for converting domain name to IP addresses so that data can be loaded from the website host.
    - When you type google.com in your browser, The DNS oversees transforming the domain, also known as the website or webpage, to the IP addresses. A DNS query is the act of inputting the domain name, and DNS resolve is the process of locating the associated IP address.
  - What are these record types? Write one line each:
    - A, AAA, CNAME, MX, NS
    - A (Address Record): This record maps a domain name to an IPv4 address. For example, if you have google.com. The A record would point to an IP 198.163.216.24. It's essential for directing web traffic to the correct server.
    - AAAA (Adress Record): Similar to the A record, but it maps a domain name to an IPv6 address. With the growing adoption of IPv6, AAAA records are used to support the larger address space.
    - CNAME: (Canonical Name Record): This record creates an alias, allowing one domain name to point to another for instance, www.google.com can be CNAME Pointing to google.com. It is useful for managing multiple service under single domain without duplicating A records.
    - MX: (Mail Exchange Record): MX record specifies the mail servers responsible for accepting email messages on behalf of a domain. They include a priority value (lower numbers have

higher priority), so if you have mail1.example.com with priority 10 and mail2.example.com with priority 20, the first server will be tried first.

- ➢ Run: dig google.com - identify the A record and TTL from the output
  - The A record for google.com is 142.250.206.174
  - TTL (Time to Live): The TTL For this A record is 240 seconds.

```
ubuntu@ip-172-31-47-149:~$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 2079
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             240     IN      A       142.250.206.174

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue Feb 17 07:21:42 UTC 2026
;; MSG SIZE  rcvd: 55
```

❖ Taks 2: IP Addressing

- ➢ What is an IPv4 address? How is it structured? (192.168.1.10)
  - IP stand for Internet Protocol version v4 stands for version four (IPv4), An IPv4 address consists of a series of four eight-bit binary numbers which are separated by decimal point.
  - It consists of 32 bits, grouped into four sections of known as octets or bytes. Each octet has 8 bits, and these bits can be represented only in 0 or 1 form.
  - It can represent 256 numbers ranging from 0 to 255. These four octets are represented as decimal numbers, separated by periods known as dotted decimals notation.

- Difference between public and private IPs – give one example of each.

Srno	Public IP	Private IP
1	Public IPs are visible to the outside world. It is like the address for your house it allows strangers to find you.	Private IPs are similar, but they are used for your private network. Only visible to you and members of your network. They are different to avoid issues with routing. What if two houses had the same address? Which one, if any, would you get your package?
2	The public IP is issued by your ISP to your modem/router, so it can connect to their network.	The private IP issued by the router for devices connecting to your personal networking.

Eg: The device that connects the two networks, usually a router, will do NAT (Network Address Translation) and should have a firewall to protect your private network from the public network.

The service that issues IPs is called DHCP. The ISP will have one and hour router will have one for your private network.

- What are the private IP ranges?
  - These private address ranges enable devices on the private network to communicate effectively without conflicting with the usage of the public IP addresses, hence the need for it in network management and security.
  - 10.x.x.x, 172.16.x.x -172.31.x.x, 192.168.x.x
  - The reserved address ranges for different classes are as follows:
    - Class A – 10.0.0.0 to 10.255.255.255
    - Class B – 172 .16.0.0 to 172.31.255.255
    - Class C – 192.168.0.0 to 192.168.255.255
- Run ip addr show – identify which of you IPs are private
  - 172.31.47.149/20 to 172.31.47.255

- 172.17.0.1/16 to 172.17.255.255
- Above range is the private IP from Class B.

```
ubuntu@ip-172-31-47-149:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 02:f2:08:39:16:73 brd ff:ff:ff:ff:ff:ff
    inet 172.31.47.149/20 metric 100 brd 172.31.47.255 scope global dynamic ens5
       valid_lft 2183sec preferred_lft 2183sec
    inet6 fe80::f2:8ff:fe39:1673/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 96:65:bf:97:60:18 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
```

❖ Task 3: CIDR & Subnetting

- What does /24 mean in 192.168.1.0/24?
  - The slash value (eg./24) shows how many bits belong to the network prefix; the rest identify individual hosts.
- How many usable hosts are in /24? A /16? A/28?
  - The usable host are in /24 is 254
  - The usable host are in A/16 is 65,534
  - The usable host are in A/28 is 14
- Explain in your own words: Why do we subnet?
  - A subnetwork is a smaller network inside a larger network. Subnetting makes networks much more efficient. Subnetting occurs when a larger network is split into smaller networks to keep security.
  - As a result, lesser networks are easier to maintain. If we examine a class A address, the potential number of hosts for each network is 224.
  - Maintaining such a larger number of hosts is challenging, but it is much simpler to maintain if we split the network into tiny sections.
- Quick exercise – fill in:

CIDR	Subnet Mask	Total IPs	Usable Hosts
/24	255.255.255.0	256	254
/16	255.255.0.0	655536	65534
/28	255.255.255.240	16	14

❖ Task 4: Port- The Door to Services

- What is a port? Why do we need them?
  - A port is a logical identifier used to distinguish different applications or services on device, allowing network traffic to reach the correct program.
  - Port works at the Transport layer, using TCP and UDP to send and receive data between devices.
  - They enable multiple applications to use the network simultaneously without interference.
  - Ports help the operating system route incoming data to appropriate applications.
- Document these common ports:

Port	Service
22	SSH (Secure Shell)
80	HTTP (Hypertext Transfer Protocol)
443	HTTPS (Hypertext Transfer Protocol Secure)
53	DNS (Domain Name System)
3306	MySQL (MY Structured Query Language)
6379	Redis
27017	Default port for MongoDB instances.

- Run ss –tulpn – match at least 2 listening ports to their services.
  - The listening Ports are 80 and 22 (HTTP) & SSH (Secure Shell)

```
ubuntu@ip-172-31-47-149:~$ ss -tulpn
Netid State   Recv-Q Send-Q       Local Address:Port     Peer Address:
Port                              Process
udp   UNCONN  0      0               127.0.0.54:53           0.0.0.0:
*
udp   UNCONN  0      0            127.0.0.53%lo:53           0.0.0.0:
*
udp   UNCONN  0      0      172.31.47.149%ens5:68            0.0.0.0:
*
udp   UNCONN  0      0                127.0.0.1:323          0.0.0.0:
*
udp   UNCONN  0      0                    [::1]:323             [::]:
*
tcp   LISTEN  0      4096            127.0.0.54:53           0.0.0.0:
*
tcp   LISTEN  0      4096         127.0.0.53%lo:53           0.0.0.0:
*
tcp   LISTEN  0      511                0.0.0.0:80           0.0.0.0:
*
tcp   LISTEN  0      4096               0.0.0.0:22           0.0.0.0:
*
tcp   LISTEN  0      4096             127.0.0.1:42563        0.0.0.0:
*
tcp   LISTEN  0      511                   [::]:80              [::]:
*
tcp   LISTEN  0      4096                  [::]:22              [::]:
*
```

- Task 5: Putting It Together
  - Answer in 2-3 lines each
  - You run curl http://myapp.com:8080- what networking concepts from today are involved?
  - Concept involved:
    - DNS resolves myapp.com to IP
    - TCP connects to port 8080
    - The HTTP request is sent to the server.
  - Your app can't reach a database at 10.0.1.50:3306- what would you check first?
    - Things to check first:
    - Check the database running?
    - Check if port 3306 is open accepting the connections

- Check if your network connectivity (Ping or telnet)

What I learned (3 key points)

1. DNS converts domain name into IP address.
2. Subnetting divides networks for better performance and security.
3. Ports allow multiple services to run on a single IP.